

V4.2 Toy Model 5E

Cosmic Coupling C(a), Horizon Scale, and Dark Energy Bridge

Rev A - Dark Sector / Unified Local-Global Capacity Test

Date: 2026.05.26 | Continuation after Toy Model 5D

Status: Toy Model 5E tests whether the same universal capacity scale A_R used for the galaxy/BTF analogue can also act as a global constant background stress producing dark-energy-like behavior. It is a dimensionless structural toy model, not a fit to cosmological data and not a derivation of Λ , H_0 , dark matter, dark energy, or physical 3T.

1. Purpose

Toy Model 5D linked the local BTF threshold to a cosmic capacity scale:

$$A_R = \xi \, c \, H_C = \xi \, c^2 / R_H$$

Toy Model 5E asks whether this same scale can play two roles: local relational capacity for galaxies and global background pressure for cosmic expansion.

2. Horizon/capacity setup

The toy expansion model uses:

$$H_{\text{eff}}^2(a) = H_0^2 [\Omega_{m0} a^{-3} + \Omega_{C0}]$$

The capacity scale is tied to the residual horizon/capacity rate:

$$H_C = H_0 \sqrt{\Omega_{C0}}$$

$$A_R = \xi \, c \, H_C$$

$$R_H = c / H_C$$

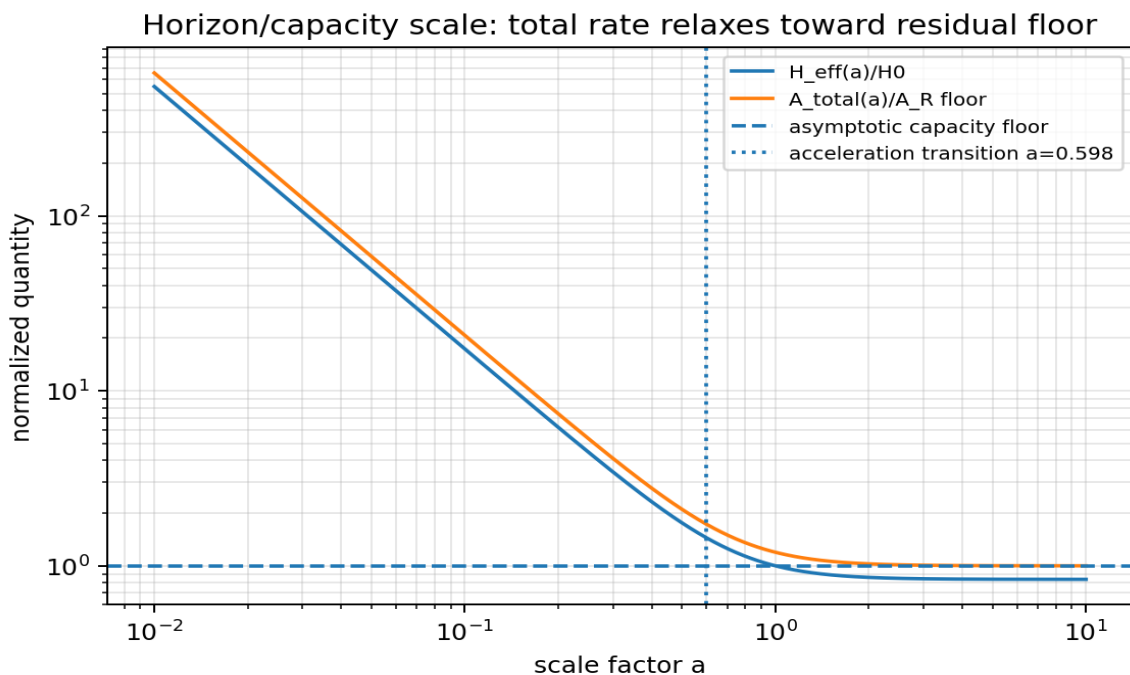
For continuity with 5D, A_R is kept at 0.3025 in dimensionless units. The factor ξ is therefore not derived; it is the factor required to match that threshold in this normalized model.

3. Capacity fraction $C_{\text{cap}}(a)$

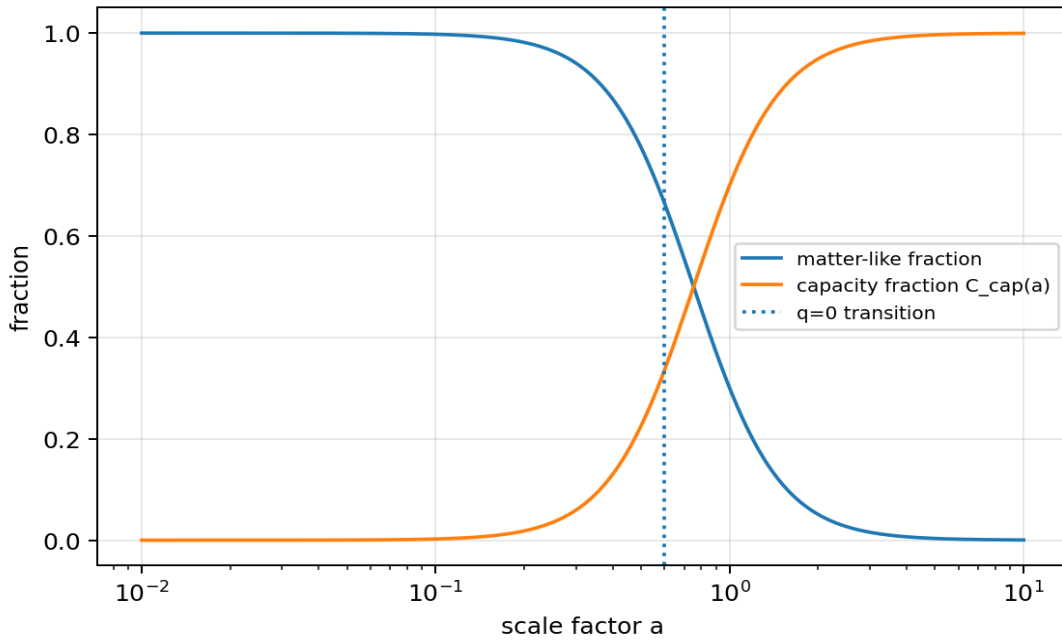
The function used in 5E is the capacity fraction:

$$\begin{aligned} C_{\text{cap}}(a) &= \Omega_C(a) \\ &= \Omega_{C0} / [\Omega_{m0} a^{-3} + \Omega_{C0}] \end{aligned}$$

This is not claimed to be the complete V4.2 coupling function $C(a)$. It is the dark-sector capacity fraction used for this test.



Capacity fraction grows as matter dilution exposes the residual background



4. Dark-energy-like background pressure

If the global capacity density is constant, then its pressure follows:

$$\rho_C = \text{constant}$$

$$E_C = \rho_C V$$

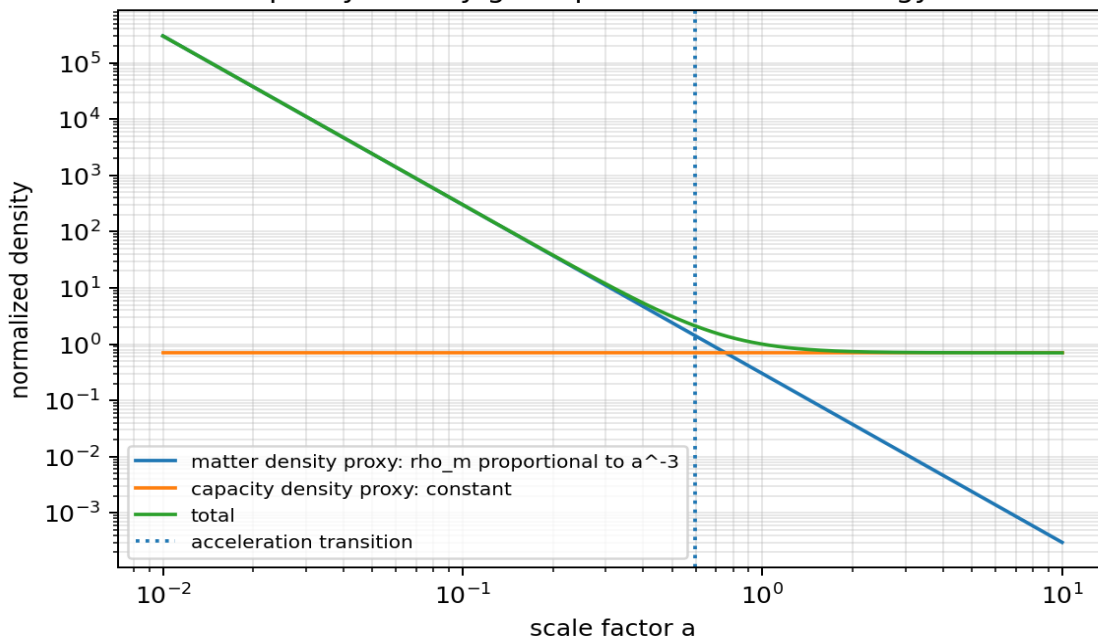
$$p_C = -dE_C/dV = -\rho_C$$

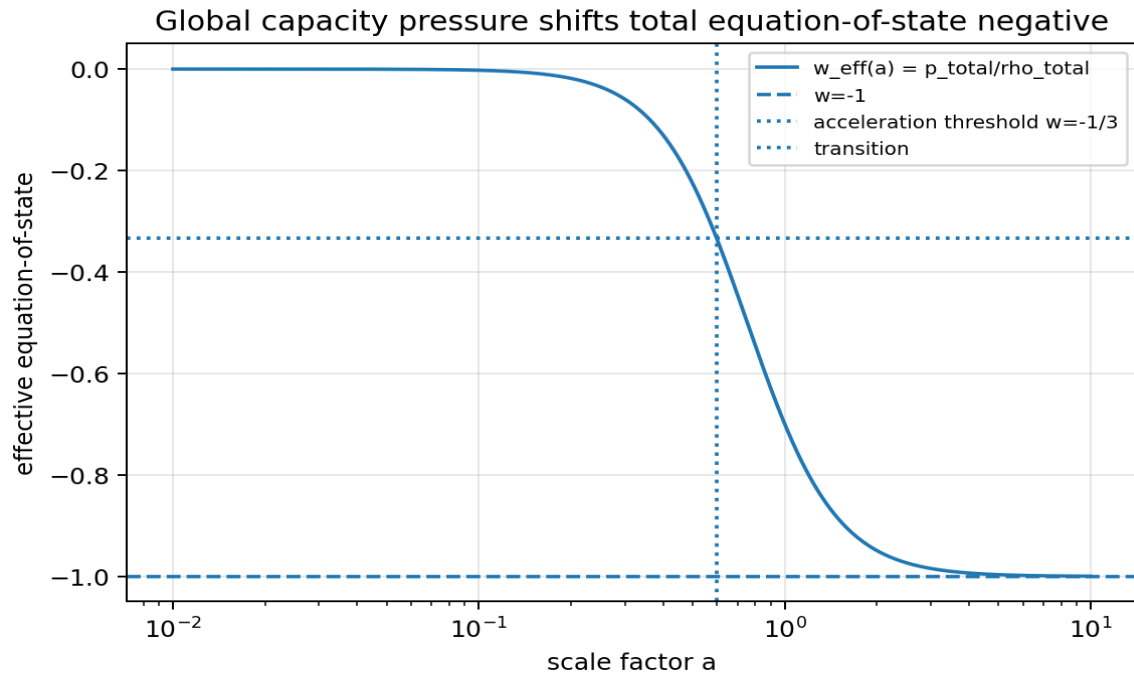
therefore:

$$w_C = p_C/\rho_C = -1$$

This is the cosmological-constant-like part of the toy model. It does not derive the physical vacuum energy; it only shows that a constant global capacity density has the correct pressure sign and equation-of-state form.

Constant capacity density gives $p=-\rho$ and dark-energy-like behavior



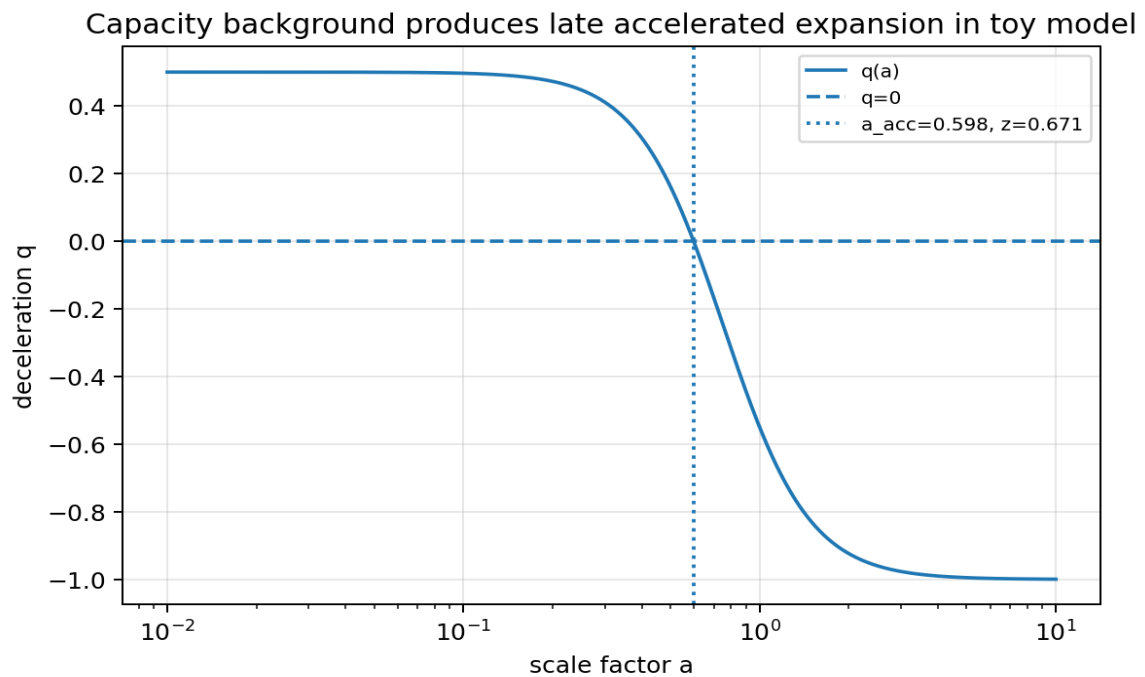


5. Acceleration condition

For a matter plus capacity background, the toy deceleration parameter is:

$$q(a) = 0.5 \Omega_{\text{m}}(a) - \Omega_{\text{C}}(a)$$

Acceleration starts when q becomes negative, i.e. when the capacity background becomes large enough relative to matter.



Diagnostic	Value	Interpretation
$\Omega_{\text{m}0}$	0.300000	Illustrative normalized matter fraction.
$\Omega_{\text{C}0}$	0.700000	Illustrative normalized capacity fraction.
H_{C}	0.836660	Residual capacity expansion scale.
A_{R}	0.302500	5D local/global threshold retained.
ξ required	0.361557	Not derived; required by chosen normalization.
R_{H} floor	1.195229	Capacity horizon length in toy units.
a_{acc}	0.598408	Scale factor where $q=0$.
z_{acc}	0.671099	Equivalent redshift in the toy normalization.
q today	-0.549986	Negative means accelerating in toy model.

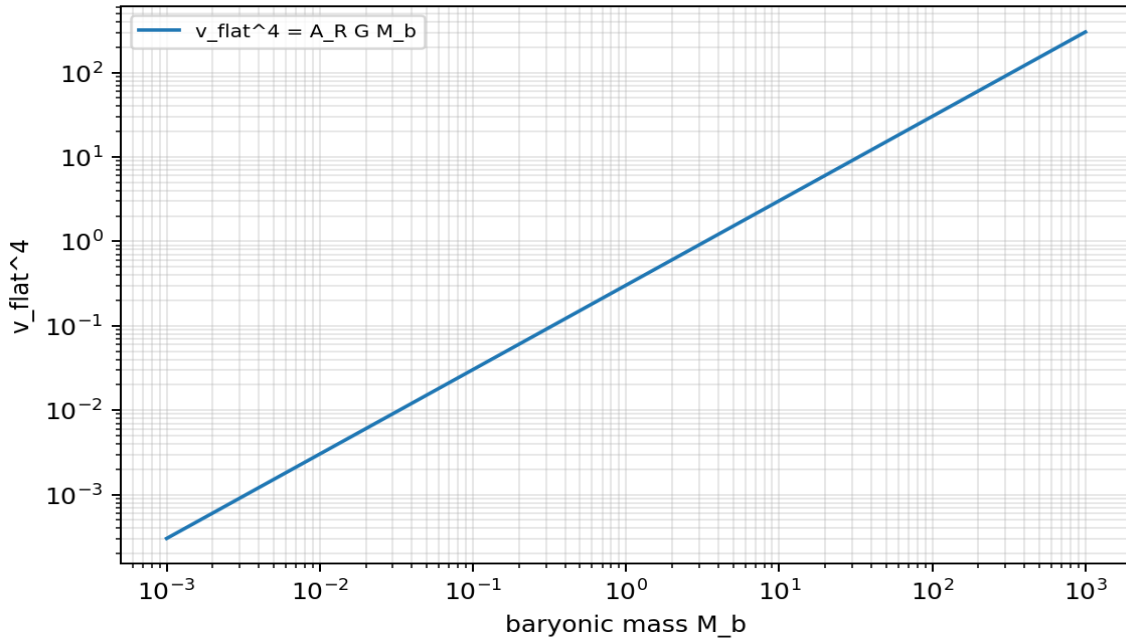
Diagnostic	Value	Interpretation
p_C + rho_C max	0.000000e+00	Checks p=-rho.

6. Unified local-global role of A_R

The same scale A_R now appears in two places:

Regime	Scale	Equation	Observable analogue	Interpretation
Local / galaxy	A_R	$T_{\text{edge}}^2 = A_R G M_b$	$v_{\text{flat}}^4 = A_R G M_b$	dark-matter-like rotation support
Global / cosmos	$A_R = \xi c H_C$	rho_C proportional to H_C^2/G	$p_C = -\rho_C$	dark-energy-like background pressure

Same A_R used locally for BTF-like scaling



7. Epoch drift guardrail

If the local threshold used the total $H_{\text{eff}}(a)$ rather than the residual capacity floor H_C , the BTF zero-point would drift with epoch. This is a potential future observational constraint.

a	z	H(a)	A_total(a)	C_cap(a)	v drift if total A used
0.25	3.000	4.460942	1.612883	0.035176	1.519565
0.50	1.000	1.760682	0.636586	0.225806	1.204434
1.00	0.000	1.000000	0.361557	0.700000	1.045593
2.00	-0.500	0.858778	0.310497	0.949153	1.006545
5.00	-0.800	0.838093	0.303018	0.996583	1.000428

5E therefore keeps the local galaxy threshold tied to the residual floor $A_R = \xi c H_C$, not the full time-varying $H_{\text{eff}}(a)$.

8. Interpretation inside V4.2

Toy Model 5E supports this limited ladder:

```

universal capacity scale A_R
-> local galaxy threshold:
    v_flat^4 = A_R G M_b

same A_R
-> residual cosmic capacity:
    H_C = A_R / (xi c)
    rho_C proportional to H_C^2/G
    p_C = -rho_C

therefore:
local dark-matter-like rotation support
and global dark-energy-like pressure
can be written from one shared capacity scale

```

It does not support this stronger ladder:

A_R physically derived
 -> Lambda measured exactly
 -> Hubble tension solved
 -> CMB and structure formation reproduced
 -> dark matter and dark energy fully explained
 -> physical 3T proven

9. Numerical checks

Check	Value	Meaning
capacity/BTF slope	1.000000	Local BTF-like scaling retained.
A _{floor} error	0.000000e+00	A _R = xi c H _C holds numerically.
rho _C from A _R	0.700000	Matches chosen capacity density fraction.
Lambda proxy	2.100000	Toy Lambda-like scale 3H _C ² /c ² .
early H slope	-1.499463	Matter-era H approx a ^{-3/2} .
late H slope	-0.012102	Late H tends constant.
early A _{total} slope	-1.499463	Capacity rate tracks H early.
late A _{total} slope	-0.012102	Capacity rate tends floor late.
w _C error	0.000000e+00	Checks w _C =-1.

10. What Toy Model 5E achieves

- It links the 5D local threshold A_R to a residual cosmic horizon/capacity scale.
- It shows that the same A_R can be written as local BTF threshold and global capacity background.
- It shows that a constant capacity density has p=-rho and w=-1 in the toy model.
- It shows late acceleration emerges once the capacity fraction dominates matter dilution.
- It identifies a future test: whether the BTF zero-point is tied to the residual floor or to the time-varying H_{eff}(a).

11. What Toy Model 5E does not achieve

- It does not derive xi.
- It does not derive the observed physical value of A_R.
- It does not derive the physical cosmological constant.
- It does not resolve the Hubble tension.
- It does not fit supernovae, BAO, CMB, or structure-growth data.
- It does not solve galaxy-cluster lensing or Bullet Cluster behavior.
- It does not disprove particle dark matter.
- It does not prove physical 3T.

12. Close-out statement for 5E

Toy Model 5E is a structural bridge between the local and global dark-sector analogues. Locally, A_R sets the galaxy-scale edge-tension threshold needed for BTF-like scaling. Globally, the same A_R can be written as a residual cosmic capacity scale, A_R = xi c H_C, whose constant density gives p=-rho and produces dark-energy-like accelerated expansion in the toy model. The result is not a cosmological solution. It is a compatibility result showing that one universal relational capacity scale can generate both local dark-matter-like and global dark-energy-like signatures at the level of controlled equations.

13. Recommended next chapter

Recommended next: Toy Model 5F - Dark Sector Track Close-Out and Observational Guardrail Map.

- Consolidate 5A-5E before attempting real data.
- Separate derived pieces from assumed pieces.
- List the required observational tests: galaxy rotation curves, BTF scatter, lensing, clusters, CMB, structure growth, and redshift drift of the BTF zero-point.
- Only after the close-out should the model attempt real galaxy datasets.

14. Rebind prompt for continuation

We are resuming after Toy Model 5E.

5B:

Derived flat rotation shape from effective 2D annular graph action.

5C:

BTF requires:

$$T_{\text{edge}}^2 = A_R G M_b$$

$$v_{\text{flat}}^4 = A_R G M_b$$

5D:

Capacity action:

$$L(T; M_b) = T^2 / (2 A_R) - G M_b \ln(T/T_{\text{ref}})$$

Variation:

$$T^2 = A_R G M_b$$

A_R structurally linked to cosmic capacity:

$$A_R = \xi c H_{\text{eff}} \text{ or } \xi c^2 / R_H$$

5E:

Separated total $H_{\text{eff}}(a)$ from residual capacity floor H_C .

Toy expansion:

$$H_{\text{eff}}^2(a) = H_0^2 [\Omega_{m0} a^{-3} + \Omega_{C0}]$$

Residual capacity:

$$H_C = H_0 \sqrt{\Omega_{C0}}$$

$$A_R = \xi c H_C$$

$$R_H = c / H_C$$

Capacity fraction:

$$C_{\text{cap}}(a) = \Omega_{C0} / [\Omega_{m0} a^{-3} + \Omega_{C0}]$$

Global capacity density:

$$\rho_C = \text{constant}$$

$$p_C = -\rho_C$$

$$w_C = -1$$

Acceleration:

$$q(a) = 0.5 \Omega_m(a) - \Omega_C(a)$$

Result:

The same A_R can act locally as the BTF threshold and globally as a dark-energy-like capacity background.

Guardrail:

ξ and physical A_R are not derived.

No real cosmology data, CMB, BAO, supernovae, structure formation, lensing, clusters, Bullet Cluster, or physical 3T proof.

Recommended next:

Toy Model 5F - Dark Sector Track Close-Out and Observational Guardrail Map.